

E-Commerce Logistics Review

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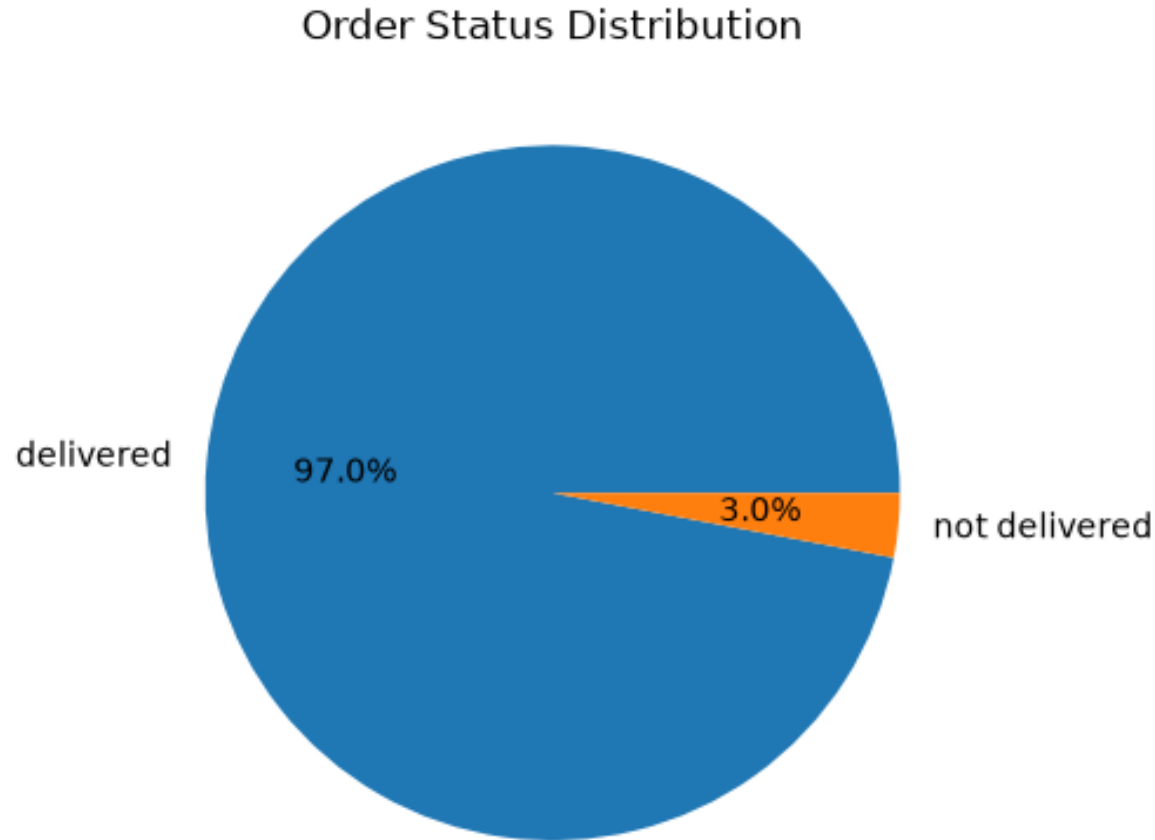
Introduction

- This project analyzes e-commerce delivery performance using the Olist Brazilian E-Commerce dataset.
- The goal is to determine whether delivery delays are concentrated in specific regions and whether late deliveries are contributing to lower customer satisfaction.
- The analysis combines **delivery performance, geographic patterns, and customer reviews** to identify key logistics issues and areas for improvement.

Summary

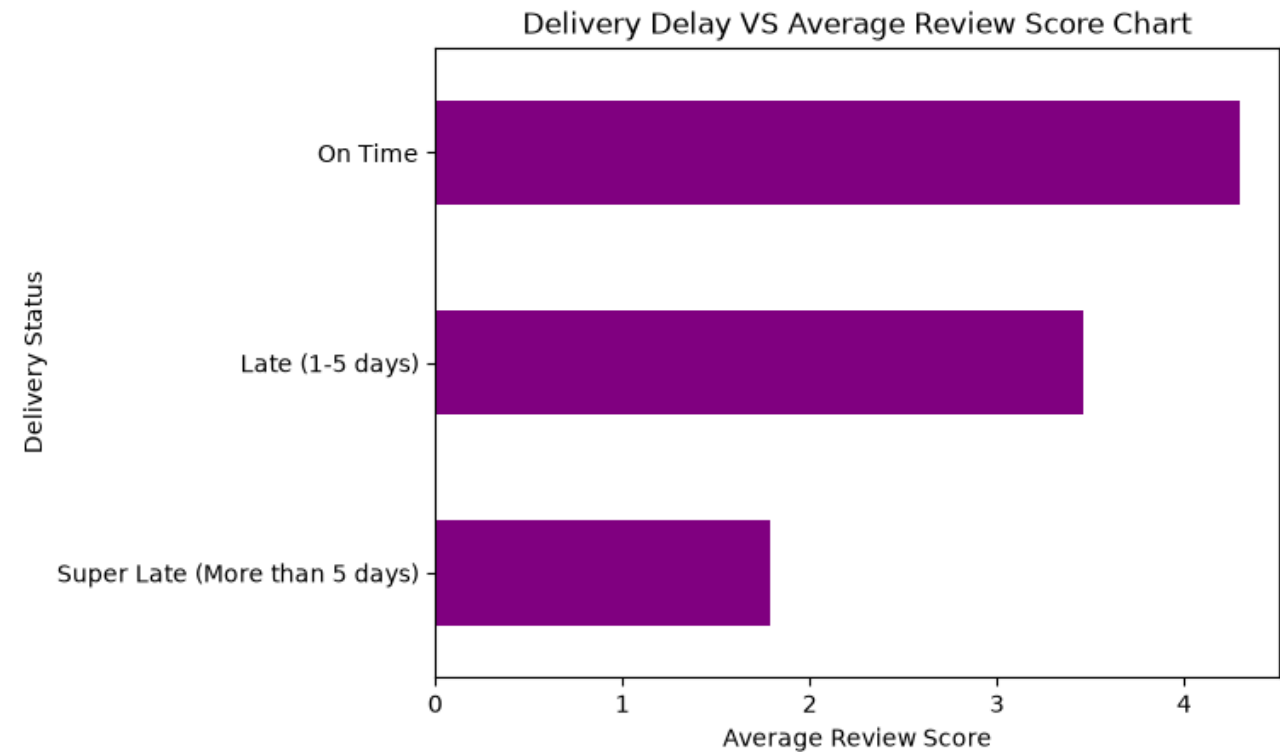
- Delivery reliability is generally high, but regional disparities exist.
- Northern states show some of the highest late-delivery percentages.
- Late deliveries are strongly associated with lower customer satisfaction, with severe delays producing the lowest ratings.

97% of Orders Reached Delivered Status



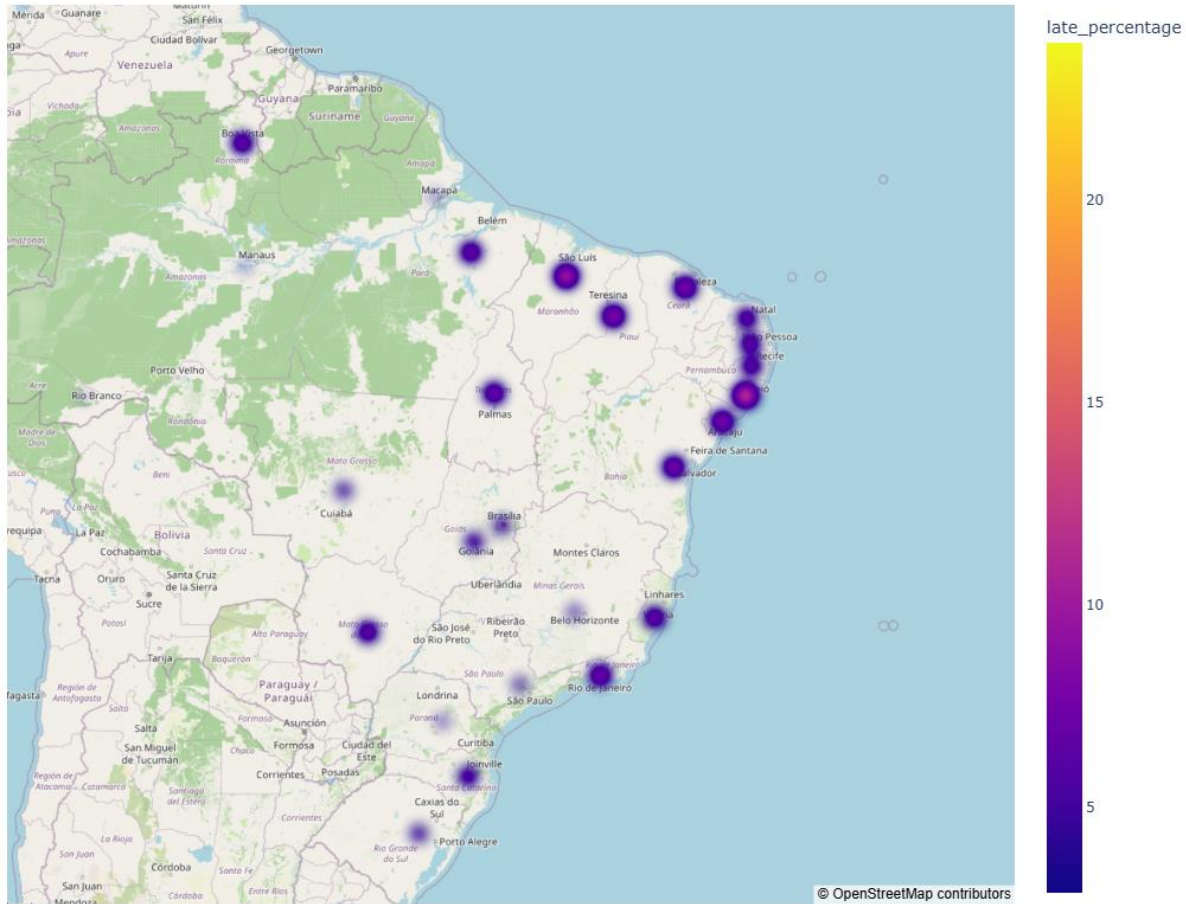
- The vast majority of orders reached the delivered stage.
- Approximately 3% remained in other fulfillment statuses, including canceled, unavailable, shipped, processing, invoiced, created and approved.

Does Late Delivery Hurt Customers?



- Not all delays are equal.
- Late deliveries resulted in reduced customer satisfaction on average.
- The satisfaction was even lower for deliveries that were late by more than 5 days.

Is It a Regional Problem?



- The far North/Northeast shows a stronger concentration of late deliveries than many southern and central states.

Recommendations

- Focus operational investigation on states with the highest late-delivery rates, particularly in the North/Northeast.
- Investigate whether estimated delivery dates are systematically too optimistic for high-risk regions.
- Track **Super Late (>5 days)** deliveries separately because they have the strongest relationship with poor customer ratings.